

SMART HEALTHCARE & TRIAGE PLATFORM

Project Overview

Integrated healthcare accessibility portal offering preliminary symptom-guided analysis, provider registrations, and emergency ambulance dispatch.

Problem Statement:	Suburban areas face delayed preliminary healthcare advice, causing critical time loss during medical emergencies.
Solution Implemented:	Developed a full-stack platform featuring a Random Forest symptom classifier, patient/doctor portals, and emergency location dispatch.
My Technical Role:	Full-Stack & Machine Learning Lead

Main Features & Technical Stack

Core Features:

- Interactive symptom triage checker (educational guidance)
- Patient & healthcare provider registration portals
- Emergency ambulance dispatcher mockup with location routing
- Medication directory and health precaution library

Technologies Used:

React, FastAPI, Python, Scikit-Learn, Random Forest, SQLite/PostgreSQL, Render

Architecture Flow:

React Frontend → FastAPI Backend → Scikit-Learn Classifier → Healthcare DB → Emergency Dispatcher

Technical Challenge & Results

Key Engineering Challenge: Integrated prominent medical disclaimers on all outputs and built standardized symptom encoding pipelines.

Validated Results: Delivered rapid symptom triage and emergency dispatch workflow coordination.

Live Demo: <https://github.com/prashant-singh-78/medicure.git> | **GitHub:** <https://github.com/prashant-singh-78/medicure.git> | **Portfolio:** prashant08.vercel.app